



**PLUMBING
& HEATING
SERVICES**

DEMINERALISING HEATING SYSTEMS

**PURER WATER MEANS GREATER
ENERGY EFFICIENCY AND
REDUCED MAINTENANCE COSTS**



**MAINTAINING YOUR
HEATING SYSTEM
IS CRUCIAL FOR ITS
EFFICIENCY AND
LONGEVITY.**

While modern heating systems tend to be more efficient and economical to run, they are also more sensitive to the quality of the heating water. Poor quality water can lead to:

- **Energy efficiency losses of 10 - 20% after the first year**
- **Increased emissions from boiler, negatively impacting the environment**
- **Increase in running and maintenance costs**
- **Increase in chemical use**

One key maintenance task is DEMINERALISING the water in your heating system.
Demineralising is a chemical free solution to treating the water that fills your heating system.

HERE'S WHY DEMINERALISATION IS IMPORTANT

Prevention of limescale deposits

According to VDI 2035, high mineral content in water, especially calcium and magnesium, leads to limescale formation. Limescale can clog pipes, reduce heat transfer efficiency, and increase energy consumption.

Reduction of Corrosion

Minerals in the water increase its electrical conductivity, which accelerates corrosion processes. Using demineralised water significantly reduces the risk of corrosion, extending the life of your heating system components.

Compliance with Standards

Following VDI 2035 guidelines not only ensures optimal system performance but is often required to maintain warranty and guarantee claims. This compliance helps avoid costly repairs and replacements due to scale and corrosion damage.

HEATING WATER PROPERLY TREATED

Hyper filtration with mobile full desalination and a filter membrane.



STEPS TO PROPERLY DEMINERALISE YOUR HEATING SYSTEM

Initial Analysis

We measure the electric conductivity, pH value, and total hardness of your heating water. If the parameters exceed the recommended values then demineralisation is necessary.

Demineralisation

Fill the system with demineralised water, either by using a demineralisation cartridge or a refill station. This step is crucial whether it is a new system or after cleaning and flushing an existing system.

Cleaning the System

For existing systems, it's essential to clean out any sludge and magnetite deposits. This can involve circulating a cleaning agent through the system for a week or multiple flushes until the water runs clear.

Regulation and Top Up

Regularly check and top up the system with demineralised water to maintain optimal water quality. Installing a heating water regulator can help manage the system's pH and prevent new deposits.

Power Flushing

Powerflushing involves flushing each section of the heating system individually. This step ensures that all contaminants are removed before filling the system with demineralised water.

Final Analysis

After filling, measure the water parameters again to ensure they meet the required standards. Continue to monitor the pH and conductivity every six months to ensure the system remains in optimal condition.

FOLLOWING THESE STEPS ENSURES YOUR HEATING SYSTEM OPERATES EFFICIENTLY, REDUCES MAINTENANCE COSTS, AND EXTENDS ITS LIFESPAN.

MAXIMUM RESULTS WITH MINIMAL DISRUPTION

To power flush and demineralise your heating system, TMG Plumbing & Heating Services use machines that work on the CIP (Cleaning In Place) principal. This works with minimal disruption while preventing further blockages and build up, future proofing your heating system.

If you would like to contact **TMG Plumbing & Heating Services** to enquire about power flushing and demineralising your heating system, contact us via our website or call **051 577089**.



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